Data augmentation

A nice party trick

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What is data augmentation?



- Feed to your network “new” training data, derived algorithmically
- Deep neural network are always data-hungry
- No data sample is completely “used”
- Computers are stupid



What is data augmentation?



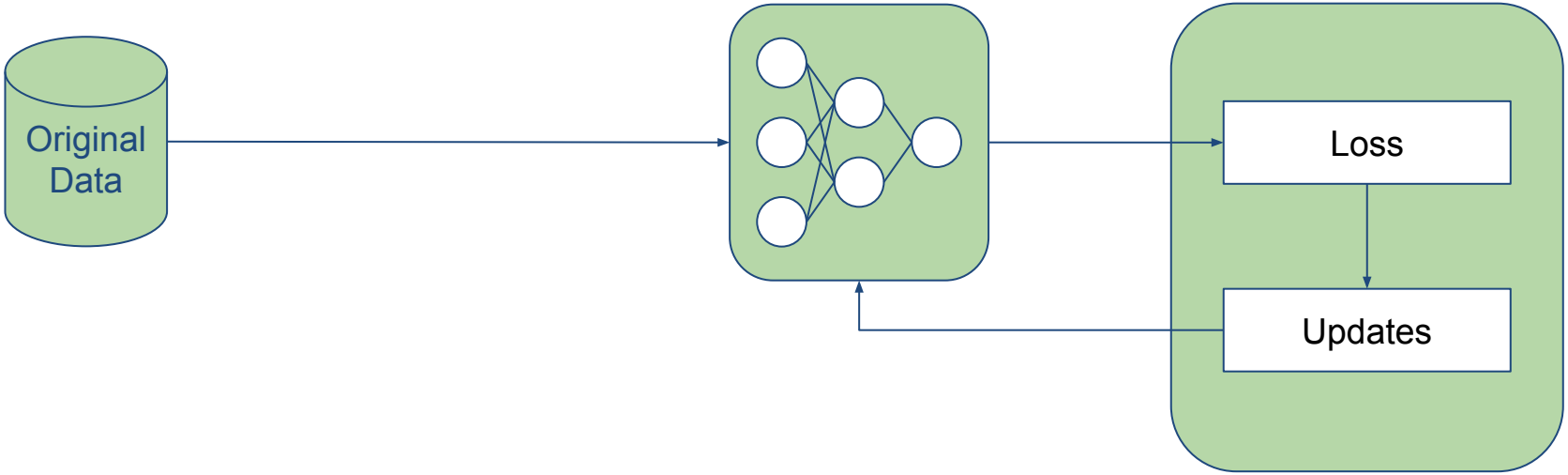
A cat



A completely different cat

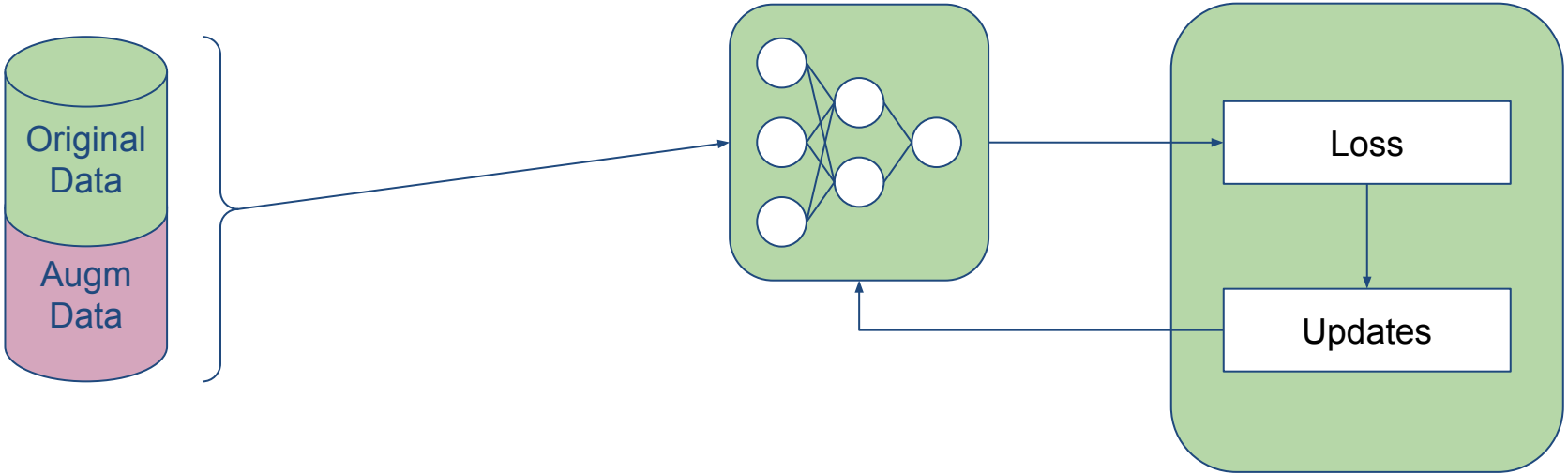


Data augmentation - training baseline



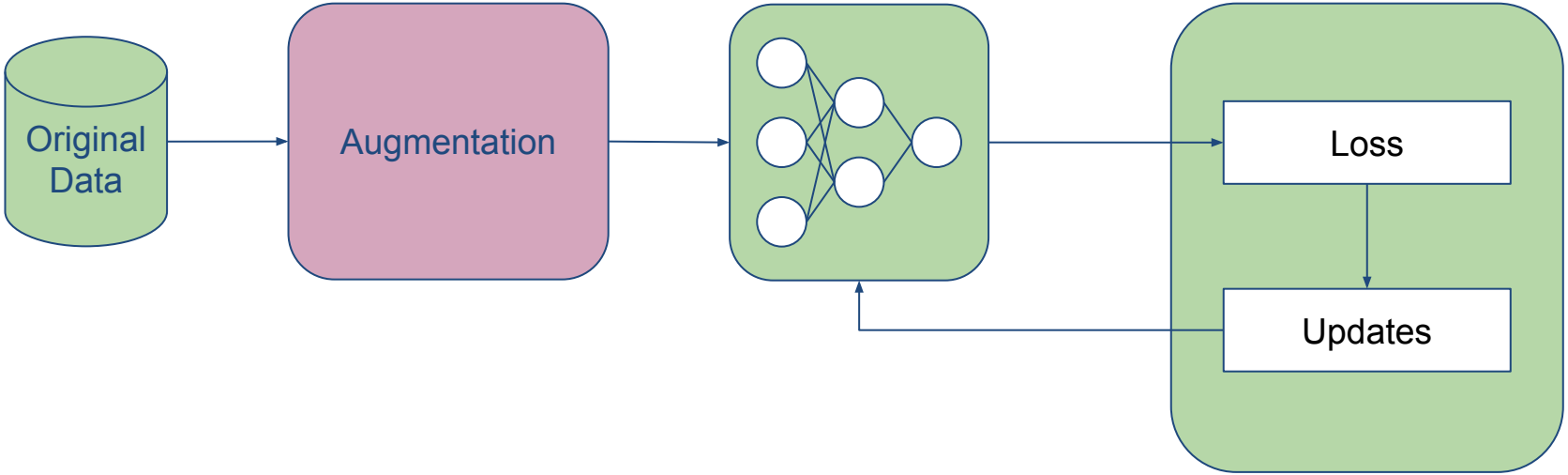


Data augmentation #1: offline



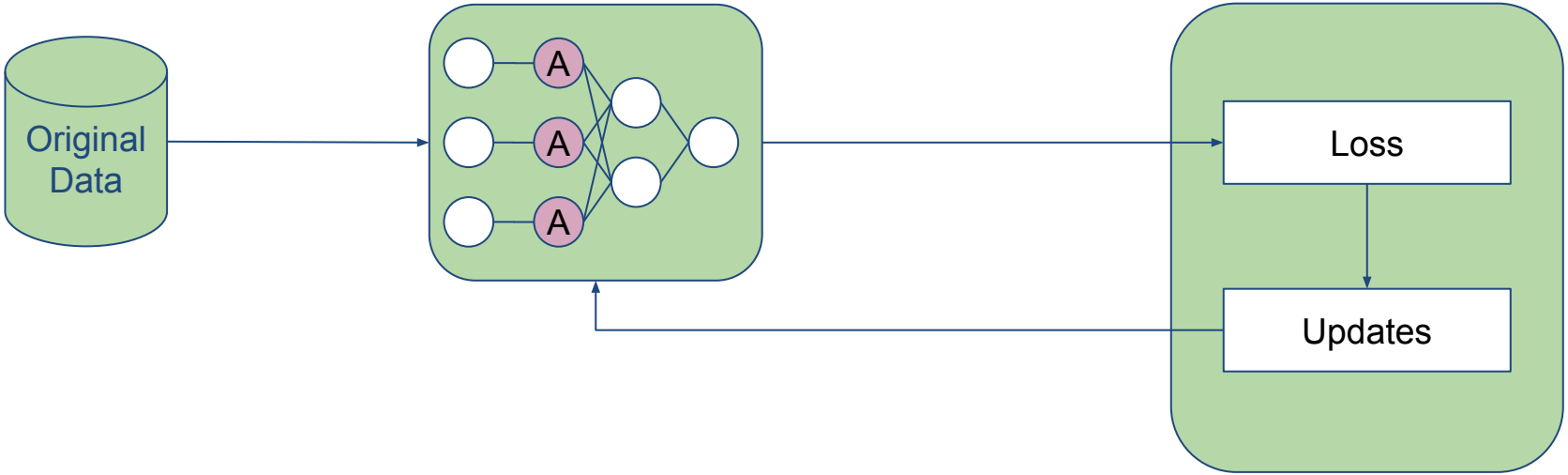


Data augmentation #2: on the fly





Data augmentation #2: on the fly (the modern keras approach)





Data augmentation effect





Credit: https://www.pyimagesearch.com/2019/07/08/keras-imagedatagenerator-and-data-augmentation/



On-the-fly data augmentation in keras

1) Organize your data properly

```
my_big_data_folder
    train_set
        class_1
            Img1.jpg Img2.jpg Img3.jpg
        class_2
            Img4.jpg Img5.jpg Img6.jpg
        class_3
            Img7.jpg Img8.jpg Img9.jpg
        ...
    test_set
        <same structure, different images>
    val_set
        <same structure, different images>
```



Data augmentation in keras (old)

2) Instantiate two/three ImageDataGenerator

```
from keras.preprocessing.image import ImageDataGenerator

train_datagen = ImageDataGenerator(
      rescale=1./255,
      horizontal_flip=True, vertical_flip=True,
      rotation_range=10, width_shift_range=0.2, height_shift_range=0.2,
      ...
)

val_datagen = ImageDataGenerator(rescale=1./255)
```



Data augmentation in keras (old)

3) Give the data to the generator

```
train_generator = train_datagen.flow_from_directory(
      directory = 'my_big_data_folder/train_set',
      target_size = image_shape,
      batch_size = batch_size,
      class_mode = 'categorical'
      ...
)

val_generator = val_datagen.flow_from_directory(
      directory = 'my_big_data_folder/val_set',
      target_size = image_shape,
      batch_size = 5,     #ATTENTION HERE
      class_mode = 'categorical'
      ---
)
```



Data augmentation in keras (old)

4) Train the model

```
history = model.fit(
      x = train_generator,
      validation_data = val_generator,
      epochs = 50,
      ...
)
```



Data augmentation in keras (new)

Instantiate one (or more) data augmentation layer(s)

```
from keras.layers import Input, Resizing, Rescaling
from keras.models import Sequential
from tf.keras.layers import RandomFlip, RandomBrightness

model = Sequential()
model.add(Input(some_input_shape))
model.add(Resizing(NEW_IMG_HEIGTH, NEW_IMAGE_WIDTH))
model.add(Rescaling(1.0/255))
model.add(RandomFlip("horizontal"))
model.add(RandomBrightness(0.5))
model.add(...) #the actual network
```



Data augmentation in keras

- Not only from directory:
  - `<your_generator>.flow_from_dataframe(...)`
- Not only images...
  - `from keras.preprocessing.sequence import TimeseriesGenerator`
  - `keras.preprocessing.text...`
- ...but images have way more options



Take home message

- Data augmentation is “free”
  - Extra computational burden is usually minimal
- It does NOT increase the training data size
  - Unless you explicitly do so (offline vs on-the-fly)
- It helps your network to generalize better
- Allows for more training epochs
- The “new” approach is more streamlined
  - But less manipulations are built-in
- It’s almost always a good idea



[REF]

- Keras image data preprocessing: https://keras.io/api/preprocessing/image/
- Adding a data augmentation layer https://www.tensorflow.org/tutorials/images/data_augmentation
- The different kinds of data augmentation, implemented in a detailed example: https://www.pyimagesearch.com/2019/07/08/keras-imagedatagenerator-and-data-augmentation/
- A gallery of image augmentation: https://machinelearningmastery.com/how-to-configure-image-data-augmentation-when-training-deep-learning-neural-networks/

